MERIDIAN PRECISION SYSTEMS
Quality and Engineering Issue Resolution

AS-IS INVESTIGATION
Prepared by StatusNeo

DETECT	TRIAGE	GATHER EVIDENCE	DIAGNOSE	DECIDE / APPROVE	CAPA
0.5 day	0.5 day	2.0 days	1.0 day	2.0 days	62 days
illustrative / simulated	illustrative / simulated	illustrative / simulated	illustrative / simulated	illustrative / simulated	illustrative / simulated

Operator

Alarm noticed
MX-400 etch hits a limit
Notes chamber and recipe

Technician

MANUAL 1 · ERROR RISK 1
Told in person
Repeats the chamber
Nothing is written yet

MANUAL 2 · ERROR RISK 2
Types the ticket
Tool, time, and symptom
Email is the record

MANUAL 4 · ERROR RISK 4
MES spreadsheet
Runs, lots, and recipe
Exported, then emailed

Process engineer

MANUAL 5 · ERROR RISK 5
Historian, SPC, ECN
Three files, copied out
Change log checked by eye

MANUAL 8 · ERROR RISK 8
Emails and a meeting
Spreadsheets compared live
No single evidence pack

MANUAL 9 · ERROR RISK 9
Tribal hypothesis
Last alarm was the chamber
Stated from memory

Quality engineer

MANUAL 3 · ERROR RISK 3
Assigns an owner
Reply-all on the mail
Recipe step often dropped

MANUAL 6 · ERROR RISK 6
Prior cases
Pulled from a folder
Forwarded into the thread

MANUAL 10
Asks for approval
Email to management
No evidence pack attached

MANUAL 12 · ERROR RISK 11
CAPA written by hand
Re-keyed into the QMS
Narrative, lot, and actions

Supply chain

MANUAL 7 · ERROR RISK 7
Supplier lots
Looked up, then emailed
Not joined to the run

Management

MANUAL 11 · ERROR RISK 10
Approves by reply
The meeting is the signature
The cause is not cited

MANUAL 13
Signs the CAPA
Paper or a workflow click
Does not see the files

ERROR RISK callouts

13 manual hand-offs. 11 places a fact can change. Illustrative / simulated.
Detect through Diagnose is 4.0 days to a hypothesis, simulated. CAPA at 62 days is the record cycle, simulated.

E1 Spoken alarm. The chamber is written from memory.
E2 Ticket retyped. Tool, time, and symptom change.
E3 Data gap. The reply drops the recipe step.
E4 Run sheet pasted into a spreadsheet.
E5 Missed engineering change. The log is checked by eye.
E6 Data gap. Prior cases sit in a folder.
E7 Wrong lot traced. The supplier lot is not joined to the run.
E8 Data gap. The meeting does not record which file was current.
E9 Bias toward the last known cause.
E10 Uncited approval. The meeting has no evidence pack.
E11 Re-keyed record. The files stay in email.

Glossary

Short forms on this sheet, spelled out.

MES manufacturing execution system
SPC statistical process control
ECN engineering change notice
QMS quality management system
CAPA corrective and preventive action

LEGEND

Operator. Notices the alarm on the tool.
Technician. The call, the ticket, the run sheet.
Process engineer. Files, the meeting, the hypothesis.
Quality engineer. Owner, the thread, the record.
Supply chain. Supplier lots by email.
Management. Approves and signs.
Red MANUAL badge and red arrow. A hand-off.
Amber ERROR RISK badge. A fact can change.

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As-is investigation flow
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Prepared by StatusNeo
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